Another derivation of Fundamental solution. (From scaling symmetry).

Find $u(t,x)$ solving $\begin{cases} \partial_t u = \partial_{xx} u \\ u|_{t=0} = \delta(x) \end{cases}$.

<u>Symmetry</u>: e.g if u is a solution, then

$u(\lambda^2 t, \lambda x)$ is also a solution.

We seek a solution $u(x,t)$ invariant under

$$u(x,t) \longmapsto \lambda^\alpha u(\lambda^\beta x, \lambda t) =: u_\lambda.$$

$$u(x,t) = \frac{1}{t^\alpha} v\left(\frac{x}{t^\beta}\right), \quad v(y) = u(y,1)$$

$$\partial_t u = -\alpha t^{-\alpha-1} v\left(\frac{x}{t^\beta}\right) + t^{-\alpha} \cdot \left(\frac{x}{t^\beta} \cdot -\beta t^{-1}\right) v'\left(\frac{x}{t^\beta}\right)$$

$$\partial_{xx} u = t^{-\alpha} \cdot t^{-2\beta} v''\left(\frac{x}{t^\beta}\right)$$

$\partial_t u = \partial_{xx} u$: all powers in t must agree: $\alpha + 1 = \alpha + 2\beta$

$\Rightarrow \beta = \frac{1}{2}$.

$u_\lambda(t,x)$ has initial condition: $u_\lambda(0,x) = \lambda^\alpha \delta(\lambda^{\frac{1}{2}} x) = \delta(x)$.

$\Rightarrow \alpha = \frac{1}{2}$ since $\int \delta(x)\,dx = 1$

So v solves $-\alpha v(r) - \beta r v'(r) = v''(r)$

$\Rightarrow \quad \frac{1}{2} v + \frac{1}{2} r v' + v'' = 0$

$\Rightarrow \quad \frac{1}{2} r v + v' = \text{const} = 0$ (const $= 0$ since $\lim_{|r| \to +\infty} v(r), v'(r) = 0$)

$\Rightarrow \quad v' = -\frac{1}{2} r v$

$\Rightarrow \quad v = C e^{-\frac{r^2}{4}}$.

Understand $u(t,x) = \int G_t(x-y)\,\phi(y)\,dy$.

① Fund. soln

Let $\Gamma(t,x;s,y) = G_{t-s}(x-y)$.

For fixed (s,y), $\Gamma(\cdot,\cdot;s,y)$ solves

$$\begin{cases} \partial_t \Gamma - \Delta \Gamma = 0, & t>s,\ x\in\mathbb{R}. \\ \lim_{t\downarrow s} \Gamma(t,x;s,y) = \delta(x-y) \end{cases}$$

i.e., the HE solution of placing a heat source at location y at time s

If initial condition is $\phi(x) = \int \delta(x-y)\,\phi(y)\,dy$

("linear combination" of δ-function)

then solution is "linear combination" of $\Gamma(t,x;s,y)$

$\Rightarrow u(t,x) = \int \Gamma(t,x;s,y)\,\phi(y)$.

(Principle of superposition)

Also, $(\mathcal{F}\delta)^\wedge(\xi) = \int \delta(x)\, e^{-2\pi i x\cdot\xi}\,dx$

$= 1$

and $\hat{G}_t(\xi) = e^{-t4\pi^2\xi^2}$.

$\Gamma(t,x;s,y)$ is called the Fundamental solution.

② Infinite speed on propagation:

$\forall \phi \geq 0$, $G_t * \phi > 0$, $\forall t>0$. (say $\phi(x) = \mathbb{1}_{[-1,1]}(x)$)

③ smoothing effect:

more decay in $\hat{f}$ $\Rightarrow$ differentiability in f.

$f(x) = \int \hat{f}(\xi)\, e^{2\pi i \xi\cdot x}\,d\xi$

Heat operator:

$\hat{\phi}(\xi) \longmapsto e^{-t4\pi^2\xi^2}\hat{\phi}(\xi)$

Non-homogeneous Problem

$$\begin{cases} \partial_t u = \Delta u + f & t>0,\ x\in\mathbb{R}^d. \\ u\big|_{t=0} = 0 \end{cases} \qquad \text{(NHE)}$$

Fundamental solut.

Recall from linear ODE:

$$\dot{x}(t) = A x(t) \implies x(t) = e^{At} x_0 =: \Phi(t) x_0$$

To solve $\dot{x}(t) = A x(t) + f(t)$, use <u>variation of constant</u>:

$$x(t) = \Phi(t)\, c(t) \qquad d\Phi(t) = A\Phi = \Phi A$$

$$\Rightarrow \dot{x}(t) = \dot{\Phi}(t)\, c(t) + \Phi(t)\, \dot{c}(t)$$

$$= A\Phi(t)\, c(t) + \Phi(t)\, \dot{c}(t)$$

$$\Rightarrow \dot{c}(t) = [\Phi(t)]^{-1} f(t) = \Phi(-t) f(t)$$

$$\Rightarrow c(t) = \int_0^t \Phi(-s) f(s)\, ds + c(0)$$

Hence, $x(t) = \Phi(t) x_0 + \Phi(t)\left[\int_0^t \Phi(-s) f(s)\, ds\right]$

$$= \Phi(t) x_0 + \int_0^t \Phi(t-s) f(s)\, ds$$

$$= \Phi(t) x_0 + \int_0^t e^{(t-s)A} f(s)\, ds.$$

Duhamel's principle

Thm Let $f \in C^{1,2}((0,\infty)\times\mathbb{R}^d) \cap C_c([0,\infty)\times\mathbb{R}^d)$

then $u(t,x) = \int_0^t \int_{\mathbb{R}^d} \Gamma(t,x;s,y) f(s,y)\, dy\, ds$ solves (NHE)

with $\lim_{(t,x)\to(0,x^0)} u(t,x) = 0$, $\forall x^0 \in \mathbb{R}^d$. $\in C^{1,2}((0,\infty)\times\mathbb{R}^d)$

Pf: $\Gamma(t,x;s,y) = G_{t-s}(x-y)$ we have

$$u(t,x) = \int_0^t \int_{\mathbb{R}^d} G_s(y)\, f(t-s, x-y)\, ds\, dy$$

∂_t, $\partial_{x_i x_j}$ on u can be passed to f

$\Rightarrow$ $u \in C^{1,2}((0,\infty)\times\mathbb{R}^d)$

(Note: $G_s(y)$ is singular near $s=0$).

$$\partial_t u - \Delta u = \int_{\mathbb{R}^d} G_t(y)\, f(0, x-y)\, dy \quad = K$$

$$+ \int_0^t \int_{\mathbb{R}} G_s(y)\, (\partial_t - \Delta_x)(f(t-s, x-y))\, ds\, dy$$

$$= K + I_1 + \int_\varepsilon^t \int_{\mathbb{R}} G_s(y)\, (-\partial_s - \Delta_y)\, f(t-s, x-y)\, ds\, dy$$

$$+ I_2 + \int_0^\varepsilon \int_{\mathbb{R}} G_s(y)\, (\partial_t - \Delta_x)\, f(t-s, x-y)\, ds\, dy$$

$$|I_2| \le \varepsilon \left(\|\partial_t f\|_{L^\infty} + \|D^2 f\|_{L^\infty} \right) \le C\varepsilon$$

$$I_1 = \int_\varepsilon^t \int_{\mathbb{R}} \left[\partial_s G_s(y) - \Delta_y G_s(y) \right] f(t-s, x-y)\, ds\, dy$$

$$- \int_{\mathbb{R}^d} G_t(y)\, f(0, x-y) + \int_{\mathbb{R}^d} G_\varepsilon(y)\, f(t-\varepsilon, x-y)\, dy$$

$$= \left(G_\varepsilon * f(t-\varepsilon, \cdot) \right)(x)$$

$$\left\| \left(G_\varepsilon * f(t,\cdot) \right) - f(t,\cdot) \right\|_{L^\infty} \to 0.$$

$$\| G_\varepsilon * f(t-\varepsilon,\cdot) - G_\varepsilon * f(t,\cdot) \|_{L^\infty} \le \| f(t-\varepsilon,\cdot) - f(t,\cdot) \|_{L^\infty} \cdot \| G_\varepsilon \|_{L^1} \to 0 \text{ as } \varepsilon \downarrow 0.$$

Hence,

$$\lim_{\varepsilon \downarrow 0} \left(G_\varepsilon * f(t-\varepsilon, \cdot) \right)(x) = f(t,x)$$

uniformly for x.

Therefore, $\partial_t u = \Delta u + f$.

<u>Last</u>: $\| u(t,\cdot) \|_{L^\infty} \leq t \| f \|_{L^\infty} \to 0$. $\boxed{\Sigma}$

<u>Use Fourier transform</u>

$$\begin{cases} \partial_t \hat{u} = -4\pi^2 \xi^2 t \cdot \hat{u}(t,\xi) + \hat{f}(t,\xi) \\ \hat{u}\big|_{t=0} = 0 \end{cases}$$

$$\Rightarrow \quad \hat{u}(t,\xi) = \int_0^t e^{-4\pi^2\xi^2(t-s)} \cdot \hat{f}(s,\xi)\, ds$$

$$\Rightarrow \quad u(t,x) = \int_0^t \left[G_{t-s} * f(s,\cdot) \right](x)\, ds$$

Let $\Gamma(t,x;s,y) = \begin{cases} G_{t-s}(y-x), & t > s \\ 0, & t \leq s \end{cases}$

$\Gamma(\cdot,\cdot;s,y)$ solves: $\begin{cases} (\partial_t - \Delta)\Gamma = \delta(t-s, x-y) \\ \Gamma(0,\cdot;s,y) = 0 \end{cases}$

So $u(t,x) = \Gamma *_{(t,x)} f$

satisfies $(\partial_t - \Delta) u(t,x) = \left(\left[(\partial_t - \Delta)\Gamma \right] *_{(t,x)} f \right)(t,x)$

$$= \int \delta(t-s, x-y)\, f(s,y) = f(t,x)$$